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Software Requirements Specification (SRS)

AI-Powered Mental Health Triage and Navigation — MVP1 Specification with MVP2+ Roadmap Annex

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Preamble

P1. Scope

This document specifies the software requirements for ClearPath, an AI-powered mental health triage and care-navigation platform. Part I (Sections 1–7) contains the normative functional and non-functional requirements for the MVP1 release. Part II (Sections 8–17) documents the architecture and design decisions that realize those requirements. Annex A describes planned future capabilities and does not constitute normative requirements for the current release.

ClearPath is scoped as a pre-clinical navigation tool, not a clinical diagnostic system. It uses validated screening instruments (PHQ-9, GAD-7) to generate a clinical profile that supports provider matching. All clinical scoring is deterministic; no clinical scores are generated by a language model.

P2. Definitions and Abbreviations

Term	Definition
PHQ-9	Patient Health Questionnaire-9. A validated 9-item self-report instrument for screening and measuring depression severity. Scores range 0–27; bands: Minimal (0–4), Mild (5–9), Moderate (10–14), Moderately Severe (15–19), Severe (20–27).
GAD-7	Generalized Anxiety Disorder 7-item scale. A validated 7-item self-report instrument for screening and measuring anxiety severity. Scores range 0–21; bands: Minimal (0–4), Mild (5–9), Moderate (10–14), Severe (15–21).
Item 9	PHQ-9 item 9 ("Thoughts that you would be better off dead, or of hurting yourself in some way"). Endorsed at any level (value ≥ 1) triggers mandatory crisis detection.

Term	Definition
Care Level	A stepped-care category derived deterministically from PHQ-9 and GAD-7 scores. Values: self_guided, coaching, therapy, psychiatry, urgent, crisis.
Deterministic	Processing that yields an identical output for a given input on every run, without language model involvement. Applied to all clinical scoring, severity banding, and care-level mapping.
LLM	Large Language Model. Used in ClearPath for three non-clinical functions: story-to-draft mapping, results summary narrative generation, and crisis-language classification on free text.
Story Path	An optional intake mode in which the user provides free-text narrative context. An LLM drafts suggested item values; the user must confirm each value before it is accepted for scoring.
Questionnaire Path	The primary intake mode. The user responds to PHQ-9 and GAD-7 items directly via a structured batched UI with 0–3 value selection.
SafetySupportCard	A UI component that surfaces crisis resources (988 Lifeline, Crisis Text Line, 911) inline when a crisis trigger is detected. Does not block intake completion.
Batch	A grouped set of PHQ-9 or GAD-7 items presented together in the questionnaire UI. The intake is structured as 7 batches across 16 items.
Session	A transient, server-side state object associating a session ID with intake answers, profile context, and processing results. No personally identifiable information is persisted to permanent storage in MVP1.
PEPM	Per-employee-per-month. The pricing unit for the B2B SaaS employer sales motion.
HPSA	Health Professional Shortage Area. A federal designation for geographic areas with insufficient mental health professional supply.
SaMD	Software as a Medical Device. ClearPath is not positioned as SaMD; it is intended for FDA General Wellness classification.
WCAG	Web Content Accessibility Guidelines. The target conformance level is WCAG 2.1 AA.
PHI	Protected Health Information as defined under HIPAA. MVP1 stores no PHI.

P3. Intended Audience

Audience	Purpose
Engineering team	Functional and non-functional requirements; API contracts; data models; acceptance criteria
Clinical reviewers	Safety requirements (§6); scoring specifications (§13.3, §13.4); crisis detection logic (§13.4)
Product and business stakeholders	System overview (§1); requirements traceability to product decisions
QA and testing	Verification and acceptance criteria (§7); testing specification (§15)

Audience	Purpose
Regulatory reviewers	Safety and compliance requirements (§6); scope definition (§1, §P1)

P4. Assumptions and Constraints

Assumptions

ID	Assumption
A-01	The user has access to a modern web browser and internet connectivity.
A-02	The user has sufficient English-language literacy to complete PHQ-9 and GAD-7 items as presented.
A-03	The user is not in acute crisis requiring immediate emergency intervention (the product addresses pre-crisis navigation, not emergency response).
A-04	Provider data in the seeded dataset is accurate at the time of seeding; real-time availability is not guaranteed.
A-05	The configured LLM provider is available and responsive for story path and summary generation. The questionnaire path functions without LLM availability.
A-06	PHQ-9 and GAD-7 scoring band definitions remain as published in the validated instrument literature; threshold tables in §13.3 reflect the current validated definitions.

Constraints

ID	Constraint
C-01	All clinical scoring (PHQ-9 total, GAD-7 total, severity banding, care-level determination) shall be implemented as deterministic logic with no LLM involvement.
C-02	LLM-suggested item values in the story path shall not be accepted for scoring without explicit user confirmation.
C-03	PHQ-9 item 9 crisis detection shall be deterministic and shall not depend on LLM availability.
C-04	The /crisis page shall function independently of the main application; it shall have no dependency on shared application state or navigation.
C-05	No personally identifiable information shall be written to permanent storage in MVP1.
C-06	API keys and all server-side clinical logic shall be inaccessible to the client.
C-07	The product shall make no diagnostic claims. All outputs shall be clearly labeled as screening results, not clinical diagnoses.
C-08	Crisis detection shall not block the user from completing screening and receiving a care plan.

Part I — MVP1: Core Requirements

1. System Overview

1.1 Product Purpose

ClearPath is a web-based mental health triage and care-navigation application. Its purpose is to reduce the time between a user's decision to seek mental health support and a qualified, actionable provider shortlist — from the current national average of approximately 48 days to minutes.

ClearPath is not a therapy platform, a diagnostic system, or a crisis intervention service. It is a navigation tool: it collects validated screening data, derives a clinical profile deterministically, and surfaces matched providers.

1.2 System Context

The system operates between a motivated user (typically an employed adult with mild-to-moderate anxiety or depression) and the fragmented U.S. mental healthcare provider network. It interposes a structured triage layer that neither the user nor the provider ecosystem currently has access to.

ClearPath is deployed as a single web application. In MVP1, it relies on a seeded provider dataset. Future phases introduce real provider data, persistent sessions, and employer-facing features.

1.3 Key System Boundaries

- ClearPath does not:
- Provide therapy or therapeutic interventions
 - Make clinical diagnoses
 - Replace crisis intervention services
 - Verify real-time provider availability
 - Access or store electronic health records
 - Transmit protected health information in MVP1

2. Functional Requirements

2.1 User Onboarding and Profile Collection

ID	Requirement
FR-001	The system shall present a landing page that states the product's purpose, scope limitations (screening, not diagnosis), and a link to the /crisis page.
FR-002	The system shall present a welcome page describing the four-step intake flow before the user begins onboarding.

ID	Requirement
FR-003	The system shall collect a user profile consisting of: ZIP code, insurance type, primary concern, and (optionally) care format preference.
FR-004	The profile wizard shall be completable in no more than three steps.
FR-005	The system shall validate that ZIP code and insurance fields are populated before proceeding to intake.
FR-006	The system shall accept an optional free-text context field during onboarding. If provided, the text shall be evaluated by the crisis classifier before intake begins.

2.2 Intake: Questionnaire Path

ID	Requirement
FR-010	The system shall present PHQ-9 and GAD-7 items using verbatim instrument wording as published in the validated instruments.
FR-011	The system shall group items into 7 batches: batches 0–2 covering PHQ-9 items 1–8 (mood chapter), batch 3 covering PHQ-9 item 9 alone (safety chapter), and batches 4–6 covering GAD-7 items 1–7 (anxiety chapter).
FR-012	Each item shall accept a user-selected response value of 0, 1, 2, or 3.
FR-013	The system shall display a progress indicator throughout the questionnaire showing current position relative to total items.
FR-014	PHQ-9 item 9 (suicidal ideation item) shall be presented as a dedicated batch (batch 3) within its own chapter.
FR-015	When PHQ-9 item 9 receives a confirmed response value of 1 or greater, the system shall display the SafetySupportCard before proceeding to the next batch.
FR-016	The SafetySupportCard shall be informational and shall not prevent the user from completing the remaining intake items.
FR-017	Upon completion of all 16 items, the system shall transition the user to the processing pipeline.

2.3 Intake: Story Path

ID	Requirement
FR-020	The system shall offer an alternative story intake path in which the user may provide a free-text narrative of at least a minimum character threshold before proceeding.
FR-021	On submission of the narrative, the system shall pass the text to the LLM story-inference service, which shall suggest a value of 0–3 for each of the 16 intake items.
FR-022	Suggested item values shall be clearly labeled as AI-generated drafts in the review interface.

ID	Requirement
FR-023	The user shall be able to review and modify each suggested value independently.
FR-024	The system shall not accept story-path item values for scoring unless the user has explicitly confirmed each item on the review screen.
FR-025	If the LLM returns a suggested value outside the range 0–3, the system shall clamp the value to the nearest valid boundary before displaying it.
FR-026	The story path shall converge with the questionnaire path at the processing stage; scoring behavior shall be identical for equivalent confirmed answers.

2.4 Scoring

ID	Requirement
FR-030	The system shall compute PHQ-9 total score as the sum of all 9 confirmed PHQ-9 item values.
FR-031	The system shall compute GAD-7 total score as the sum of all 7 confirmed GAD-7 item values.
FR-032	The system shall assign a PHQ-9 severity band according to the validated scoring thresholds: Minimal (0–4), Mild (5–9), Moderate (10–14), Moderately Severe (15–19), Severe (20–27).
FR-033	The system shall assign a GAD-7 severity band according to the validated scoring thresholds: Minimal (0–4), Mild (5–9), Moderate (10–14), Severe (15–21).
FR-034	Scoring shall be performed entirely by deterministic logic. No LLM shall be involved in score computation.
FR-035	Item values with <code>confirmed = false</code> shall be excluded from all score computations.

2.5 Care-Level Determination

ID	Requirement
FR-040	The system shall determine a care level for each completed intake using the following deterministic mapping:
FR-041	If <code>isCrisis = true</code> (PHQ-9 item 9 $\geq$ 1 or LLM classifier confidence $\geq$ 0.85 on free text), care level shall be <code>crisis</code> .
FR-042	If <code>isCrisis = false</code> and (PHQ-9 band = Severe OR GAD-7 band = Severe), care level shall be <code>urgent</code> .
FR-043	If PHQ-9 band = Moderately Severe OR GAD-7 band = Moderate (without meeting crisis or urgent conditions), care level shall be <code>psychiatry</code> .
FR-044	If PHQ-9 band = Moderate OR GAD-7 band = Moderate (without meeting higher-tier conditions), care level shall be <code>therapy</code> .

ID	Requirement
FR-045	If PHQ-9 band = Mild OR GAD-7 band = Mild (without meeting higher-tier conditions), care level shall be <code>coaching</code> .
FR-046	If PHQ-9 band = Minimal AND GAD-7 band = Minimal, care level shall be <code>self_guided</code> .
FR-047	The <code>crisis</code> care level and the <code>urgent</code> care level shall be mutually exclusive and independently defined; a user with PHQ-9 item 9 ≥ 1 shall always receive <code>crisis</code> , not <code>urgent</code> .

2.6 Crisis Detection

ID	Requirement
FR-050	The system shall implement two independent crisis detection layers: a deterministic layer (PHQ-9 item 9) and an LLM-based classifier layer (free text).
FR-051	The deterministic layer shall evaluate the confirmed value of PHQ-9 item 9. If the value is 1 or greater, the system shall set <code>isCrisis = true</code> and <code>crisis source = "item9"</code> .
FR-052	The deterministic crisis check shall not depend on LLM availability. It shall function on the questionnaire path regardless of LLM configuration.
FR-053	The LLM classifier layer shall evaluate any free-text input (optional context, story narrative). If the classifier returns a confidence score of 0.85 or greater, the system shall set the classifier crisis flag and include it in the final <code>isCrisis</code> determination.
FR-054	A <code>CrisisCheck</code> result shall include: crisis (boolean), confidence (float), and source ("item9", "classifier", or "none").
FR-055	When <code>isCrisis = true</code> at any point in the intake flow, the SafetySupportCard shall be displayed inline.
FR-056	The SafetySupportCard shall display: 988 Suicide and Crisis Lifeline, Crisis Text Line, 911, and a link to the /crisis page.
FR-057	Crisis detection shall not trigger mandatory redirection away from the intake flow. The user shall be able to complete screening and reach the results page.
FR-058	When <code>isCrisis = true</code> at results, a safety banner shall be displayed prominently on the results page alongside the care plan and provider shortlist.

2.7 Provider Matching

ID	Requirement
FR-060	The system shall filter the provider dataset to retain only providers whose insurance types include the user's stated insurance.
FR-061	The system shall filter the remaining provider set to retain providers whose specialty aligns with the user's stated concern or care level.
FR-062	The system shall apply care format filtering (in-person, telehealth, either) based on the user's stated format preference.

ID	Requirement
FR-063	The system shall rank the filtered provider set by fit score and return the top 3 matches as the provider shortlist.
FR-064	Each provider card in the results shall include: name, credentials, specialty, insurance accepted, format, estimated wait time, and at least one match reason.
FR-065	If no providers satisfy all filter criteria, the system shall present a community resources fallback rather than an empty result.
FR-066	Provider cards shall include a "verify before booking" notice and a last-updated timestamp to communicate data currency.

2.8 Results and Summary

ID	Requirement
FR-070	The results page shall display a clinical profile card summarizing PHQ-9 score and band, GAD-7 score and band, and derived care level.
FR-071	The clinical profile card shall include a clear non-diagnostic disclaimer stating that the results represent screening output, not a clinical diagnosis.
FR-072	The system shall generate a plain-language, non-diagnostic summary narrative via LLM, personalized using the user's stated concern and care level.
FR-073	The LLM prompt for summary generation shall instruct the model to: use non-clinical language, include a non-diagnostic framing statement, not suggest a specific diagnosis, and not recommend specific medications.
FR-074	If LLM summary generation fails or times out, the system shall display a deterministic fallback summary; the user shall reach the results page regardless of LLM availability.
FR-075	The results page shall display a processing pipeline visualization showing score, match, and summary steps.
FR-076	The results page shall include a print/save option that renders the care packet including clinical profile, care level, and provider shortlist.

2.9 Crisis Page

ID	Requirement
FR-080	The /crisis page shall load independently of the main application. It shall have no dependency on shared application state, session data, or navigation.
FR-081	The /crisis page shall display: 988 Suicide and Crisis Lifeline, Crisis Text Line, 911.
FR-082	The /crisis page shall function even if the main application fails to load.
FR-083	The /crisis page shall be accessible from the landing page, results page, and SafetySupportCard at all times.

2.10 Session Management

ID	Requirement
FR-090	The system shall maintain session state using a session ID generated at intake start.
FR-091	Session state shall be stored in the browser (sessionStorage) on the client and in an in-memory store on the server.
FR-092	If a user returns to the intake start with an existing session ID, the system shall resume from the current batch state rather than resetting.
FR-093	Session data shall not persist beyond the browser session (client) or server cold start (server-side in-memory store).

3. Non-Functional Requirements

3.1 Performance

ID	Requirement
NFR-001	The end-to-end intake flow (from profile completion to results page render) shall complete within a median of 7 minutes under normal usage conditions.
NFR-002	API routes for scoring and matching shall respond within 2 seconds under normal load for the MVP1 seeded dataset.
NFR-003	The /crisis page shall load and render static content within 3 seconds on a standard mobile connection.

3.2 Availability

ID	Requirement
NFR-010	The questionnaire path and deterministic scoring shall function without LLM availability. LLM-dependent features (story path, results summary) shall degrade gracefully when the LLM provider is unavailable.
NFR-011	The /crisis page shall be accessible independently of the main application's availability.

3.3 Accessibility

ID	Requirement
NFR-020	The intake form and results page shall conform to WCAG 2.1 Level AA.
NFR-021	All form elements shall have associated labels.
NFR-022	The application shall be navigable by keyboard.
NFR-023	Color is not used as the sole means of conveying information.

3.4 Responsiveness

ID	Requirement
NFR-030	All pages shall render correctly at a minimum viewport width of 375px (mobile).
NFR-031	The intake questionnaire shall be fully operable on a mobile touch interface.

3.5 Security

ID	Requirement
NFR-040	LLM API keys shall be accessible only in server-side execution contexts. They shall never be exposed to the client.
NFR-041	All clinical scoring logic shall execute server-side. The client shall not receive scoring algorithms.
NFR-042	The system shall not log or persist user-provided free text to permanent storage in MVP1.

3.6 Maintainability

ID	Requirement
NFR-050	The codebase shall have no TypeScript compilation errors at build time.
NFR-051	ESLint shall pass without warnings on core application files.
NFR-052	Scoring thresholds shall be defined in a single authoritative location; no threshold values shall be hardcoded in multiple places.

4. Data Requirements

4.1 User Profile

The user profile captures onboarding context for session threading and match filtering. It shall contain:

- ZIP code (string, required)
- Insurance type (string, required; must correspond to a valid option in the provider dataset)
- Primary concern (one of: anxiety, depression, relationship, stress, unsure)
- Care format preference (optional; one of: telehealth, in-person, either)
- Optional context text (optional free-text string; used for crisis classification and summary personalization only)

4.2 Intake Answer

Each intake answer shall record:

- Instrument identifier (PHQ9 or GAD7)

- Item index within the instrument (0-based)
- Response value (integer in {0, 1, 2, 3})
- Confirmation flag (boolean; items with `confirmed = false` are excluded from scoring)
- Optional raw text (for story-path provenance tracking only)

4.3 Score Result

The score result shall contain:

- PHQ-9 total (integer, 0–27)
- PHQ-9 severity band (one of: minimal, mild, moderate, moderately_severe, severe)
- PHQ-9 item 9 value (integer in {0, 1, 2, 3})
- GAD-7 total (integer, 0–21)
- GAD-7 severity band (one of: minimal, mild, moderate, severe)
- Care level (one of: self_guided, coaching, therapy, psychiatry, urgent, crisis)
- isCrisis (boolean)
- isUrgent (boolean; mutually exclusive with isCrisis)

4.4 Provider Record

Each provider record in the dataset shall contain:

- Unique identifier
- Name and credentials
- Specialty list (one or more)
- Accepted insurance types (one or more)
- Care format (telehealth, in-person, or either)
- ZIP code
- Estimated wait days
- Cost per session (integer or null if not available)

4.5 Match Result

The match result shall contain:

- A list of up to 3 matched provider cards, each containing the provider record, a computed fit score, and a list of match reasons
- An optional fallback list of community resources, populated when no providers satisfy filter criteria

5. Interface Requirements

5.1 User Interface

ID	Requirement
UI-001	The interface shall use plain, non-clinical language throughout the intake and results flow.
UI-002	PHQ-9 and GAD-7 items shall display the verbatim instrument text without modification.
UI-003	Non-diagnostic disclaimers shall appear at the following locations: landing page, welcome page, intake choice page, results page (adjacent to clinical profile card), and within the LLM results summary.
UI-004	LLM-generated content on the story review page shall be visually differentiated from user-confirmed values.
UI-005	The care level recommendation shall be presented as a care category name and description; it shall not use clinical diagnostic language.

5.2 External Service Interfaces

Service	Interface Type	MVP1 Usage
LLM provider (Hugging Face / OpenAI / OpenRouter)	HTTP REST API via server-side abstraction	Story mapping, results summary, crisis text classification
Vercel hosting	Deployment platform	Serverless deployment of Next.js application

No real-time provider availability APIs, insurance eligibility APIs, or EHR interfaces are in scope for MVP1.

6. Safety and Security Requirements

6.1 Clinical Safety Requirements

ID	Requirement
SS-001	The system shall never generate PHQ-9 or GAD-7 total scores using a language model. Scores shall be derived exclusively from deterministic computation on confirmed user responses.
SS-002	PHQ-9 item 9 crisis detection shall be deterministic and shall evaluate the confirmed user response directly. It shall not depend on LLM inference.
SS-003	The LLM crisis classifier shall supplement, not replace, the deterministic item 9 check.
SS-004	Care-level <code>crisis</code> shall be assigned if and only if <code>isCrisis = true</code> . <code>isCrisis = true</code> is set by item 9 ≥ 1 (deterministic) or LLM classifier confidence ≥ 0.85 .

ID	Requirement
SS-005	The care levels <code>crisis</code> and <code>urgent</code> shall be independently and non-overlappingly defined.
SS-006	Story-path intake shall not permit unconfirmed AI-suggested item values to enter the scoring computation.
SS-007	Crisis detection shall never result in mandatory redirection away from the intake flow.
SS-008	A user whose intake results in <code>isCrisis = true</code> shall receive a provider shortlist and care plan alongside crisis resources.

6.2 Prompt Security Requirements

ID	Requirement
SS-010	All LLM calls shall use a hardened system prompt that prohibits the model from: providing therapy, clinical advice, or diagnoses; generating PHQ-9 or GAD-7 numeric scores; recommending specific medications; providing specific provider recommendations.
SS-011	The system prompt shall constrain the LLM to the specific task assigned (story-item mapping, summary generation, or crisis classification).
SS-012	The system shall be tested with adversarial inputs designed to elicit clinical advice, diagnostic statements, or score generation from the LLM. No such outputs shall pass the system prompt constraint in testing.

6.3 Data Privacy Requirements

ID	Requirement
SS-020	MVP1 shall store no personally identifiable information in any persistent storage.
SS-021	Session data shall be ephemeral: stored in browser sessionStorage (client-side) and server in-memory state (server-side).
SS-022	Text provided to the LLM provider shall be limited to the minimum required for the specific API call.
SS-023	LLM API credentials shall reside in server-side environment variables only.
SS-024	No user data shall be shared with third parties other than the configured LLM provider for the specific API call.

6.4 Disclaimer Requirements

ID	Requirement
SS-030	The system shall display a non-diagnostic disclaimer at the landing page before the user enters the flow.

ID	Requirement
SS-031	The system shall display a non-diagnostic disclaimer at the welcome page.
SS-032	The system shall display a non-diagnostic disclaimer at the intake choice page before screening begins.
SS-033	The results page shall display a non-diagnostic disclaimer adjacent to the clinical profile card.
SS-034	The LLM summary prompt shall instruct the model to include a non-diagnostic framing statement in the generated narrative.

7. Verification and Acceptance Criteria

7.1 Functional Acceptance Criteria

ID	Criterion	Pass Condition
AC-001	PHQ-9 scoring accuracy	System PHQ-9 total shall match manual scoring for a minimum of 10 test cases spanning all band boundaries.
AC-002	GAD-7 scoring accuracy	System GAD-7 total shall match manual scoring for a minimum of 10 test cases spanning all band boundaries.
AC-003	Item 9 crisis detection	PHQ-9 item 9 value ≥ 1 shall result in isCrisis = true and care level = crisis in 100% of test runs.
AC-004	Crisis UX continuity	User shall be able to complete intake and reach the results page when item 9 ≥ 1 ; SafetySupportCard and results safety banner shall both be present.
AC-005	Story path scoring integrity	An intake completed via story path with confirmed answers equivalent to a questionnaire-path intake shall produce identical PHQ-9 and GAD-7 totals.
AC-006	Unconfirmed answer exclusion	A story-path intake in which item values are not confirmed on the review screen shall not result in those values contributing to the score.
AC-007	LLM score exclusion	LLM shall not produce or modify any PHQ-9 or GAD-7 total score in a minimum of 20 test runs across all intake paths.
AC-008	Intake completion time	Median completion time from profile start to results page shall be ≤ 7 minutes across a minimum of 5 test runs.
AC-009	Match relevance	Matched providers shall include only providers whose accepted insurance includes the user's stated insurance type and whose specialty is relevant to the stated concern.
AC-010	Community fallback	When no providers match the filter criteria, the community resources fallback shall be displayed.
AC-011	Crisis page independence	/crisis page shall load and display 988, Crisis Text Line, and 911 resources without requiring main application state.

7.2 Non-Functional Acceptance Criteria

ID	Criterion	Pass Condition
AC-020	Mobile responsiveness	All pages render correctly and all intake controls are operable at 375px viewport width.
AC-021	Accessibility	WCAG 2.1 AA audit of intake and results pages passes.
AC-022	Build integrity	TypeScript compilation produces no errors.
AC-023	Lint	ESLint passes with no warnings on core application files.

7.3 Safety Verification Checklist

ID	Check	Pass Condition
SV-001	Item 9 presented in dedicated safety chapter	Batch 3 contains only PHQ-9 item 9; chapter label is "safety".
SV-002	Item 9 trigger → SafetySupportCard	Item 9 response ≥ 1 causes SafetySupportCard to display before next batch; screening continues.
SV-003	Crisis intake → results with safety banner	Completing intake after item 9 trigger reaches results page with crisis care level and safety banner.
SV-004	/crisis page loads	/crisis displays 988, Crisis Text Line, 911 on direct navigation.
SV-005	High-risk story → no hard block	Story text containing crisis-adjacent language does not prevent intake completion; SafetySupportCard displays.
SV-006	Benign story → no false crisis flag	A benign story narrative does not trigger the crisis classifier.
SV-007	Adversarial prompt → no clinical advice	LLM does not return therapy, diagnostic, or score-generation content under adversarial prompt inputs.

Part II — Architecture and Design Specification

8. Architectural Overview

8.1 Summary

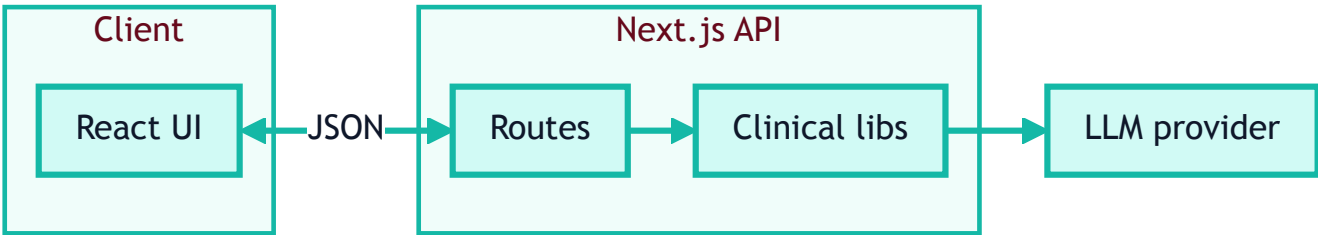
MVP1 is a single Next.js (App Router) full-stack application deployed on Vercel. The UI and all API routes are contained within a single deployable unit. A language model is used for three specific functions: optional story-to-draft screening item mapping, plain-language results summary generation, and crisis-language classification on free text.

All clinical logic — scoring, severity classification, care-level determination, provider ranking, and PHQ-9 item 9 crisis detection — is implemented as deterministic server-side logic with no language model involvement.

Architectural decisions deferred beyond MVP1:

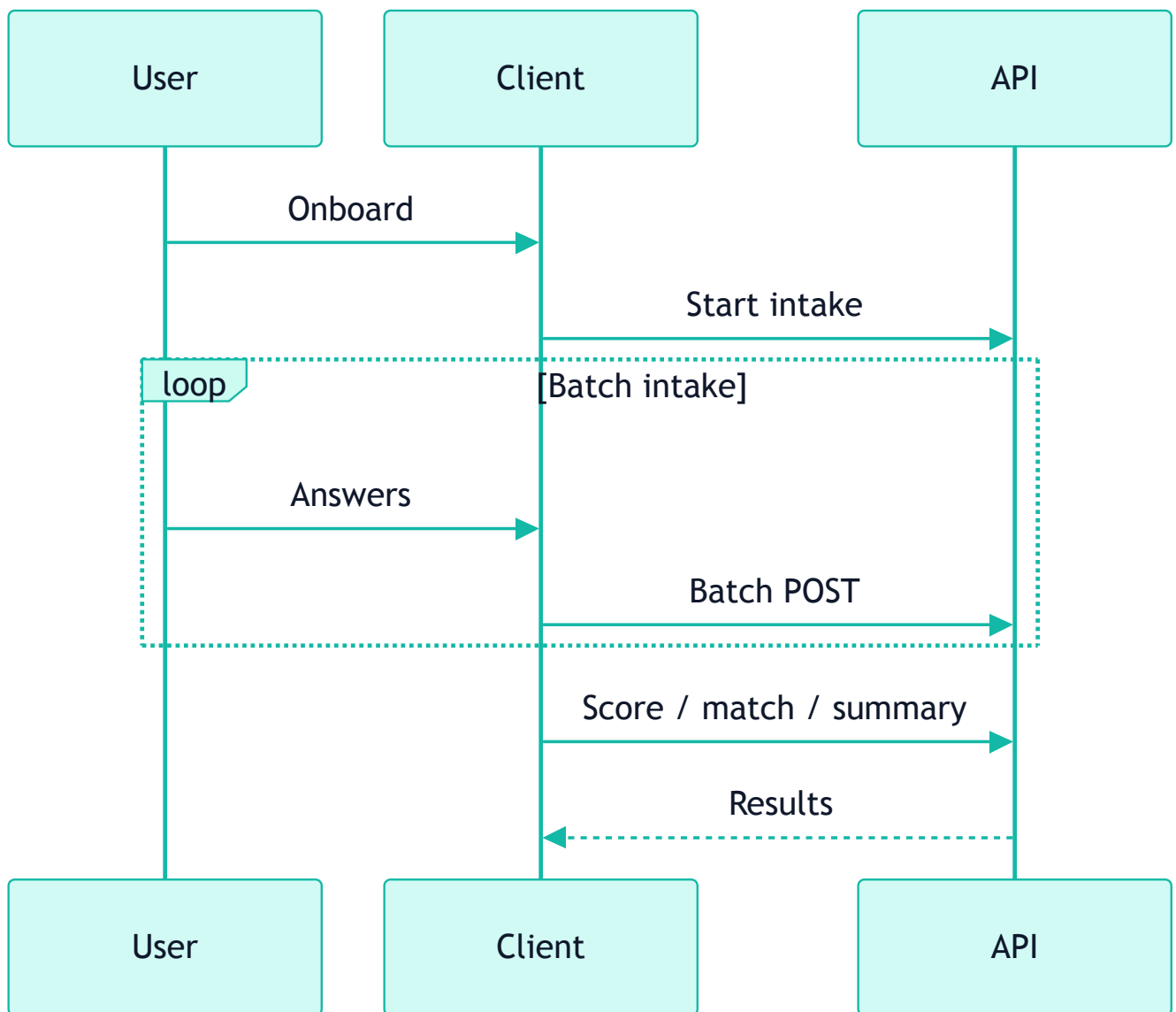
Approach	Status	Rationale
FastAPI separate service	MVP3	AI orchestration complexity does not justify a second service in MVP1
LangGraph workflow orchestration	MVP3+	Linear batch intake does not require a graph workflow in MVP1
Conversational LLM intake	Excluded	Higher latency, hallucination risk, and test surface; batched UI is more reliable
Postgres / database server	MVP2	Seeded JSON proves the match loop; persistent storage planned for MVP2
Auth / accounts	MVP2	Anonymous sessions reduce friction and privacy exposure in MVP1

8.2 Component Diagram



8.3 Request Lifecycle: Happy Paths

Standard Questionnaire Path



Optional Story Path



Crisis Resource Navigation



9. Technology Stack

Layer	Technology	Version / Notes
Framework	Next.js (App Router)	v16; TypeScript throughout

Layer	Technology	Version / Notes
Styling	Tailwind CSS	v4; theme tokens in <code>app/globals.css</code>
AI Inference	HF / OpenAI / OpenRouter	Via <code>lib/ai/openai.ts</code> ; configured by <code>LLM_*</code> env vars
Crisis Classification	Rules + LLM	<code>lib/screening/crisis.ts</code> ; LLM threshold ≥ 0.85 confidence
Clinical Logic	Pure TypeScript	<code>lib/screening/</code> , <code>lib/matching/</code> ; zero LLM involvement
Provider Data	<code>data/providers.json</code>	Seeded 20+ providers; 3 specialties, 4 insurance types, 3 ZIP clusters
Client Persistence	<code>sessionStorage</code>	<code>lib/storage/client-storage.ts</code> ; session ID, profile, results snapshot
Server Persistence	In-memory map	<code>lib/storage/session.ts</code> ; ephemeral across Vercel cold starts (MVP1 limitation)
Testing	Vitest	50 tests across scoring, matching, intake flow, care-level, story inference, LLM provider
Deployment	Vercel	https://clearpathcare-demo.vercel.app

Environment configuration:

Variable	Description
<code>LLM_PROVIDER</code>	Provider identifier (e.g., <code>huggingface</code>)
<code>LLM_API_KEY</code>	API token
<code>LLM_BASE_URL</code>	Provider base URL
<code>LLM_MODEL</code>	Model identifier

Provider-specific aliases (`HF_*`, `OPENAI_API_KEY`, `OPENROUTER_*`) are supported. The questionnaire-only path operates without LLM configuration. The story path and results summary require a configured LLM.

10. System Boundaries and Trust Model

10.1 Inside MVP1 Scope

Component	Notes
Next.js app (UI + API routes)	Single deployable on Vercel
Deterministic scoring, matching, and routing	Pure TypeScript; no external dependencies
Seeded <code>data/providers.json</code>	~20 providers for demonstration

Component	Notes
Dual-layer crisis detection	Deterministic item-9 + LLM classifier
Static <code>/crisis</code> page	Loads independently; no app dependencies
Browser sessionStorage + server in-memory session	Ephemeral; no PHI persistence

10.2 Outside MVP1 Scope (Deferred)

Component	Target Phase
OpenAI / HF API (external LLM)	Always external
NPPES / live provider availability	MVP2
Insurance eligibility APIs / clearinghouses	MVP3
Accounts, auth, HIPAA datastore	MVP2+
EHR / booking integration	Long-term
FastAPI AI service	MVP3
LangGraph workflow	MVP3+ (only if justified)

10.3 Trust Boundary

API keys and all clinical scoring logic reside server-side only. The client never:

- Computes PHQ-9 or GAD-7 scores
- Applies severity band logic
- Calls the LLM provider directly
- Handles API credentials

All scoring, matching, and AI calls are server-side API routes. The client submits answers and receives structured results.

11. Data Models

These are TypeScript transport data transfer objects (DTOs). No long-term PHI storage exists in MVP1.

11.1 Patient Profile (Onboarding Context)

The profile captures user ZIP, insurance, concern, format preference, and optional free-text context. `optionalContext` is used for summary personalization and crisis classification; it is not used for silent scoring of intake items.

11.2 Intake Answer

Each intake answer records instrument identifier (PHQ9 or GAD7), item index (0-based), response value (0–3), confirmation status (boolean), and optional raw text for story-path provenance.

11.3 Score Result

The score result records PHQ-9 total, PHQ-9 severity band, PHQ-9 item 9 value, GAD-7 total, GAD-7 severity band, derived care level, isCrisis (boolean), and isUrgent (boolean).

Scoring thresholds (deterministic):

Instrument	Band	Score Range
PHQ-9	Minimal	0–4
PHQ-9	Mild	5–9
PHQ-9	Moderate	10–14
PHQ-9	Moderately Severe	15–19
PHQ-9	Severe	20–27
GAD-7	Minimal	0–4
GAD-7	Mild	5–9
GAD-7	Moderate	10–14
GAD-7	Severe	15–21

Care-level mapping (deterministic):

Condition	Care Level
isCrisis = true (item 9 $\geq$ 1 or classifier confidence $\geq$ 0.85)	crisis
PHQ-9 Severe or GAD-7 Severe, and isCrisis = false	urgent
PHQ-9 Moderately Severe or GAD-7 Moderate (below urgent/crisis threshold)	psychiatry
PHQ-9 Moderate or GAD-7 Moderate (below psychiatry threshold)	therapy
PHQ-9 Mild or GAD-7 Mild	coaching
PHQ-9 Minimal and GAD-7 Minimal	self_guided

11.4 Intake Batch State (API Response)

The batch state response includes: batch index, total batch count, chapter identifier and intro content, item list with global index and safety flag, progress counters, done flag, and a CrisisCheck result.

11.5 Crisis Check

The crisis check response includes: crisis (boolean), confidence (float 0–1), and source ("item9", "classifier", or "none").

11.6 Provider Record

Each provider record includes: unique ID, name, credentials, specialty list, accepted insurance list, care format, ZIP code, estimated wait days, cost per session (nullable), and avatar seed.

11.7 Match Result

The match result includes a list of matched provider cards (each with provider record, fit score, and match reasons) and an optional community resources fallback list.

12. API Specification

All routes accept `POST` requests with JSON request/response bodies. The `sessionId` field in the request body threads server-side session state.

12.1 Route Reference

Route	Purpose	Key Inputs	Key Outputs
<code>POST /api/intake/start</code>	Begin or resume batch intake; crisis-check optionalContext	<code>{ sessionId, onboarding, mode }</code>	IntakeBatchState
<code>POST /api/intake/batch</code>	Submit one batch of answers; return next batch state	<code>{ sessionId, answers[] }</code>	IntakeBatchState
<code>POST /api/intake/infer-story</code>	Classify story for crisis; LLM drafts 0–3 per item	<code>{ sessionId, story }</code>	<code>{ suggestions[], crisis }</code>
<code>POST /api/intake/review-submit</code>	Persist all 16 confirmed review answers	<code>{ sessionId, answers[] }</code>	<code>{ crisis, done }</code>
<code>POST /api/intake/session</code>	Return current session answers, story, and AI suggestions	<code>{ sessionId }</code>	<code>{ answers[], story, aiSuggestions[] }</code>
<code>POST /api/intake/score</code>	Compute deterministic ScoreResult	<code>{ sessionId }</code>	ScoreResult
<code>POST /api/match</code>	Compute deterministic MatchResult	<code>{ sessionId, scoreResult }</code>	MatchResult
<code>POST /api/summary</code>	Generate LLM non-diagnostic narrative with deterministic fallback	<code>{ sessionId, scoreResult, matchResult }</code>	<code>{ summary: string }</code>

Out of scope for MVP1: Legacy conversational intake routes are not implemented. Crisis classification is handled within `POST /api/intake/infer-story` and `POST /api/intake/start`.

12.2 Crisis Behavior at the API Layer

Trigger	API Behavior	UI Behavior
PHQ-9 item 9 $\geq$ 1	Batch response includes <code>crisis: { crisis: true, source: "item9" }</code>	SafetySupportCard shown; screening continues
Story classifier $\geq$ 0.85	<code>infer-story</code> sets classifier crisis flag on session	Flagged; affects final isCrisis at score time
Profile context classifier	<code>start</code> may set initial crisis flag	Logged; does not block flow or redirect
Final <code>isCrisis</code> at results	<code>score</code> sets <code>isCrisis = true</code>	Safety banner on /results; /crisis link offered

12.3 Error Handling

All API routes shall return structured responses with a `success` flag. On failure, the response shall include an `error` string and optional error code. Client-side: summary failures shall invoke deterministic fallback text; the user shall reach the results page regardless.

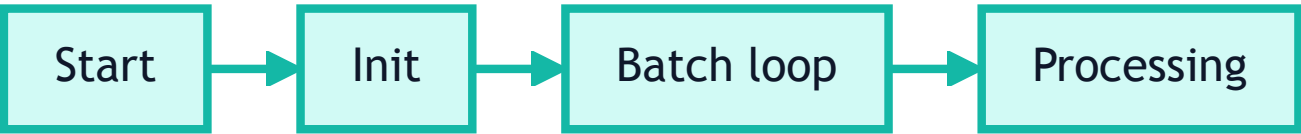
13. Core Workflow Specification

13.1 Batch Intake Engine

The intake is structured as 7 batches across 16 items (PHQ-9 items 0–8, GAD-7 items 9–15):

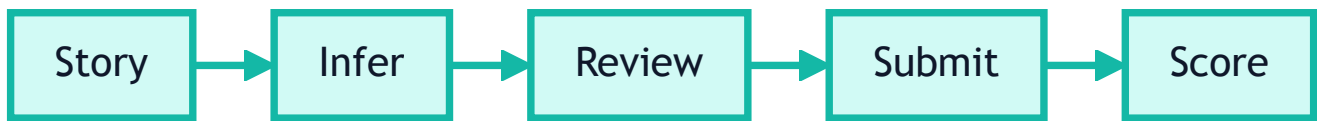
Batch	Items (global indices)	Chapter	Notes
0	0, 1, 2	mood	PHQ-9 items 1–3
1	3, 4, 5	mood	PHQ-9 items 4–6
2	6, 7	mood	PHQ-9 items 7–8
3	8	safety	PHQ-9 item 9 — alone; triggers SafetySupportCard if value $\geq$ 1
4	9, 10, 11	anxiety	GAD-7 items 1–3
5	12, 13, 14	anxiety	GAD-7 items 4–6
6	15	anxiety	GAD-7 item 7

Session flow:



Session resume: If a client boots with an existing session ID, the start endpoint shall return the current batch state rather than resetting the session.

13.2 Story Review Path



LLM story inference constraints:

- The LLM shall suggest a value of 0–3 per item; it shall not interpret clinical severity
- Suggestions shall be clearly labeled as drafts in the review UI
- The user may override any suggestion to any value 0–3
- Out-of-range LLM values shall be clamped to the nearest valid boundary before display

13.3 Scoring Engine Behavior

The scoring engine shall:

- Accept a complete set of confirmed intake answers
- Sum confirmed PHQ-9 answers to produce the PHQ-9 total
- Extract the confirmed value of PHQ-9 item 9 separately
- Sum confirmed GAD-7 answers to produce the GAD-7 total
- Derive severity bands from the totals using the thresholds in §11.3
- Derive care level from bands, item 9 value, and classifier crisis flag using the mapping in §11.3
- Set `isCrisis = true` if PHQ-9 item 9 ≥ 1 or classifier crisis flag is set
- Set `isUrgent = true` if `isCrisis` is false and PHQ-9 band is Severe or GAD-7 band is Severe

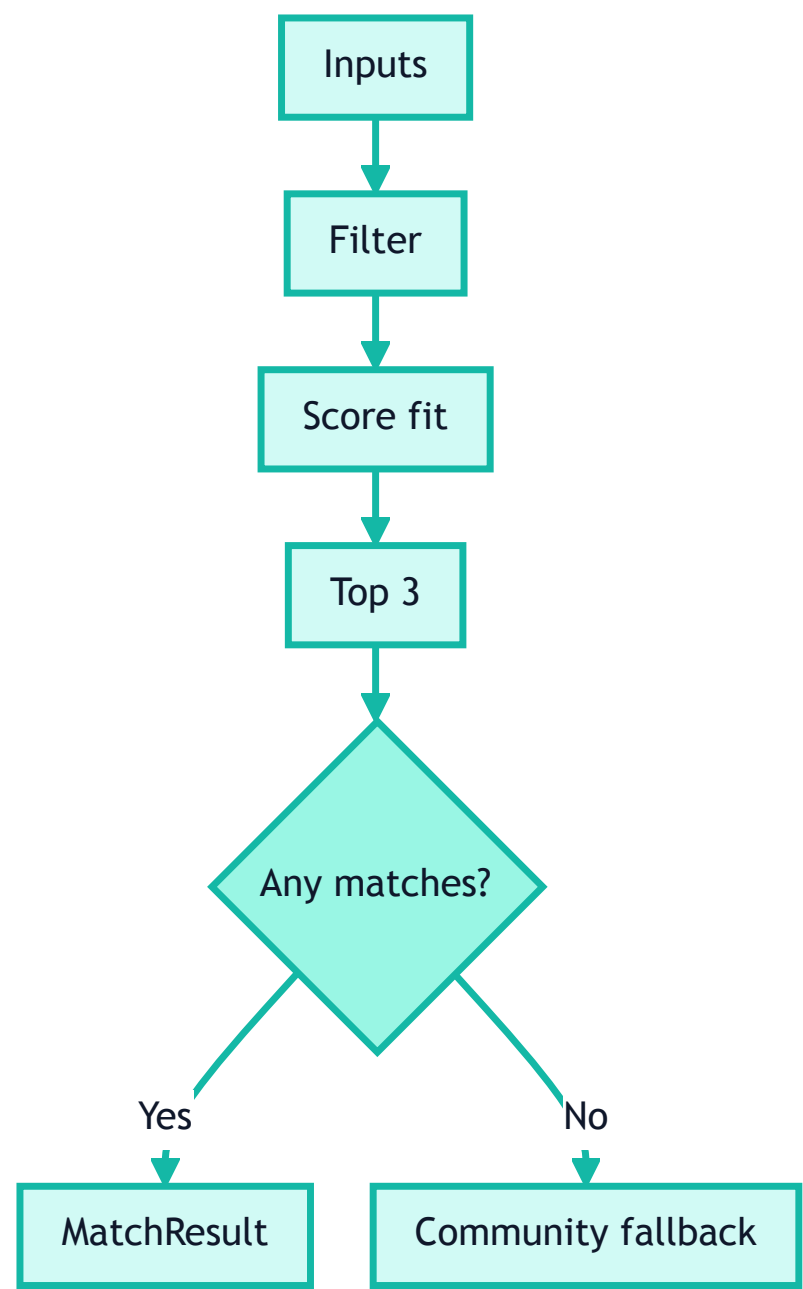
13.4 Crisis Detection Behavior

Layer 1 — Deterministic (always active): The system shall evaluate the confirmed value of PHQ-9 item 9 directly. If the value is 1 or greater, it shall return `{ crisis: true, confidence: 1.0, source: "item9" }`. This check shall not invoke the LLM under any circumstances.

Layer 2 — LLM classifier (on free text only): For story and optional context text, the system shall invoke the LLM crisis classifier with a dedicated system prompt. If the LLM returns a confidence score of 0.85 or greater, it shall return `{ crisis: true, confidence: <score>, source: "classifier" }`. This layer supplements but does not replace the deterministic layer.

Layer 3 — UX (always visible when triggered): The SafetySupportCard component shall display 988 Suicide and Crisis Lifeline, Crisis Text Line, and 911. It shall be non-blocking. The safety card shall also appear as a banner on the results page for any intake with `isCrisis = true`.

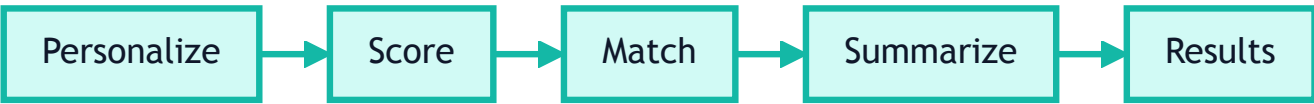
13.5 Provider Matching Behavior



The matching engine shall apply hard filters (insurance, specialty, format) before computing fit scores. It shall return the top 3 providers by fit score. If no providers pass all filters, it shall return the community resources fallback.

13.6 Client Finalization Pipeline

The processing page shall orchestrate the following sequential steps and display step-by-step progress to the user:



If the summary step fails, the pipeline shall substitute a deterministic fallback summary and complete normally.

14. Pages and Routes

Route	Component	Role	Notes
/	Landing	Marketing landing	Value prop, screening disclaimer, crisis link
/welcome	Welcome	Journey preview	Entry point to onboarding; describes the 4-step flow
/onboarding	Profile wizard	Multi-step profile	ZIP, insurance, concern, format; 3-step wizard
/intake	Intake choice	Path selection	User selects questionnaire path or story path
/intake/questions	Batched questionnaire	PHQ-9/GAD-7 UI	BatchQuestionnaire component; progress indicator; chapter intros
/intake/story	Narrative entry	Free-text intake	Character minimum; submit triggers story inference
/intake/review	AI draft review	Story review	16 items with AI suggestions; user confirms each
/processing	Processing pipeline	Pipeline progress	Score → match → summary steps
/results	Results display	Output page	ClinicalProfileCard, ProviderCard ×3, HowResultsWork, safety banner
/crisis	Crisis resources	Static crisis page	Dark theme; 988, Crisis Text Line, 911; no app dependencies

- Route protection:** No authentication in MVP1. All routes are public. Session state is managed via sessionStorage (client) and an in-memory map (server).
- Crisis page independence:** /crisis uses a dedicated layout with no shared dependencies. It is a static page that functions if the main application fails.

15. Testing Specification

15.1 Automated Test Requirements

The system shall have automated test coverage for the following domains:

Domain	Required Coverage
PHQ-9 scoring	All band boundary values (0, 4, 5, 9, 10, 14, 15, 19, 20, 27) and item 9 crisis flag

Domain	Required Coverage
GAD-7 scoring	All band boundary values (0, 4, 5, 9, 10, 14, 15, 21)
Care-level mapping	All input combinations across band and crisis flag values; urgent vs. crisis distinction; edge cases
Provider matching	Insurance filter, specialty filter, format filter, ranking logic, community fallback trigger
Intake flow	Batch progression, item 9 crisis detection, session resume, review-submit flow
Story inference	LLM draft normalization, out-of-range value clamping, missing item handling
LLM provider	Provider initialization, fallback behavior, error handling

The current test suite comprises 50 automated tests executed by Vitest. All tests shall pass as a prerequisite for production deployment.

15.2 Manual End-to-End Verification

The following personas shall be verified against the production deployment before each release:

Persona	Path	Key Verification
Mild anxiety	Questionnaire	Low severity bands; coaching or self-guided care level; provider match includes relevant specialty
Moderate depression	Questionnaire	Therapy-level care; item 9 = 0; no crisis banner
Crisis (item 9 ≥ 1)	Questionnaire	Crisis care level; safety banner on results; providers still shown; /crisis accessible
Story path	Story + review	AI draft displayed; scores computed only after confirmation; equivalent confirmed answers produce identical scores as questionnaire path

15.3 Safety Verification Requirements

All six safety verification checks in §7.3 shall be executed and documented before release.

15.4 Non-Functional Verification Requirements

Test	Condition
Responsive design	Renders correctly at 375px viewport
Accessibility	WCAG 2.1 AA audit passes on intake + results pages
Build integrity	Compilation produces no TypeScript errors
Lint	ESLint passes with no warnings on core files

16. Deployment Specification

16.1 Production Environment

Parameter	Value
Host	Vercel (hobby tier; instant deploys from GitHub main)
URL	https://clearpathcare-demo.vercel.app
Environment variables	Set in Vercel dashboard: LLM_PROVIDER , LLM_API_KEY , LLM_BASE_URL , LLM_MODEL

16.2 Post-Deploy Verification

The following checks shall be performed after every production deployment:

- 1. Confirm / , /welcome , and /crisis load successfully.
- 2. Run the questionnaire-path persona end-to-end on production.
- 3. Confirm LLM-dependent paths (story intake or results summary) respond as expected.

16.3 Known Limitations (MVP1)

- 1. **Session persistence:** In-memory sessions do not survive Vercel cold starts. Addressed in MVP2 with a persistent session store.
- 2. **Rate limiting:** API routes are not rate-limited in MVP1. Production hardening should include Vercel rate limiting or middleware if traffic increases.
- 3. **LLM availability:** If the configured LLM provider is unavailable, story mapping may fail; results summary uses a deterministic fallback; scoring and matching are unaffected on the questionnaire path.

17. Project Structure

```
app/  
  page.tsx  
  welcome/page.tsx  
  onboarding/page.tsx  
  intake/  
    page.tsx  
    questions/page.tsx  
    story/page.tsx  
    review/page.tsx  
  processing/page.tsx  
  results/page.tsx  
  crisis/  
    page.tsx  
    layout.tsx  
  api/  
    intake/  
      start/route.ts  
      batch/route.ts  
      infer-story/route.ts  
      review-submit/route.ts  
      session/route.ts  
      score/route.ts  
    match/route.ts  
    summary/route.ts  
  
lib/  
  ai/  
    openai.ts  
    prompts.ts  
    story-inference.ts  
  intake/  
    intake-flow.ts  
    instruments.ts  
    finalize-intake-client.ts  
  screening/  
    scoring.ts  
    crisis.ts  
    screening-summary.ts  
  storage/  
    session.ts  
    client-storage.ts  
  matching/  
    matcher.ts  
    provider-avatar.ts  
  hooks/  
    use-session-ready.ts  
  types.ts
```

```
validators.ts
utils.ts

components/
  shared/
    safety-support-card.tsx
  intake/
    batch-questionnaire.tsx
    chapter-intro.tsx
    processing-screen.tsx
  results/
    clinical-profile-card.tsx
    provider-card.tsx
    how-results-work.tsx

data/
  providers.json

tests/
  scoring.test.ts
  care-level.test.ts
  matcher.test.ts
  intake-flow.test.ts
  story-inference.test.ts
  llm-provider.test.ts

.env.example
README.md
```

Annex A — Future Roadmap: MVP2 and Beyond

Annex scope statement. This annex describes planned future capabilities for ClearPath MVP2 and beyond. Content in this annex represents intended future scope and does not constitute normative requirements for the MVP1 release. Architecture decisions, feature specifications, and infrastructure changes described here are subject to revision based on MVP1 validation results and product learnings.

A1. Scope Boundary: MVP1 vs. Future

Capability	MVP1	MVP2 (Days 8–30)	MVP3+ (Months 2–6)
Structured PHQ-9/GAD-7 (batched UI, verbatim)	Complete	Refine UX	Adaptive multi-instrument

Capability	MVP1	MVP2 (Days 8–30)	MVP3+ (Months 2–6)
Optional story → AI draft → user review	Complete	Clinical review of mapping	Richer narrative intake
Deterministic scoring + care level	Complete	Unchanged	Unchanged
Dual-layer crisis detection + safety card + /crisis	Complete	Hardened classifier + audit log	Continuous monitoring
Provider match	Complete (seeded)	Real data	Embeddings-based semantic match
Results + summary + processing pipeline	Complete	+ Trends	+ Shareable clinician handoff
Server session persistence	In-memory only	Postgres/session store	Full persistence
Provider availability (real-time)	Out of scope	Freshness signal	Near-real-time
Accounts / longitudinal tracking	Out of scope	Opt-in	Full
Employer dashboard	Out of scope	v0 (aggregate)	Analytics + ROI
Provider notifications / referrals	Out of scope	v1	Booking handoff
Insurance verification	Out of scope	Stub	Live eligibility
FastAPI AI service	Out of scope	Introduced	Core
LangGraph orchestration	Out of scope	Evaluate	Adopted where justified
HIPAA-compliant architecture	Out of scope (no PHI)	Designed	Enforced for pilots
Multi-language	Out of scope	Spanish	Top-N languages

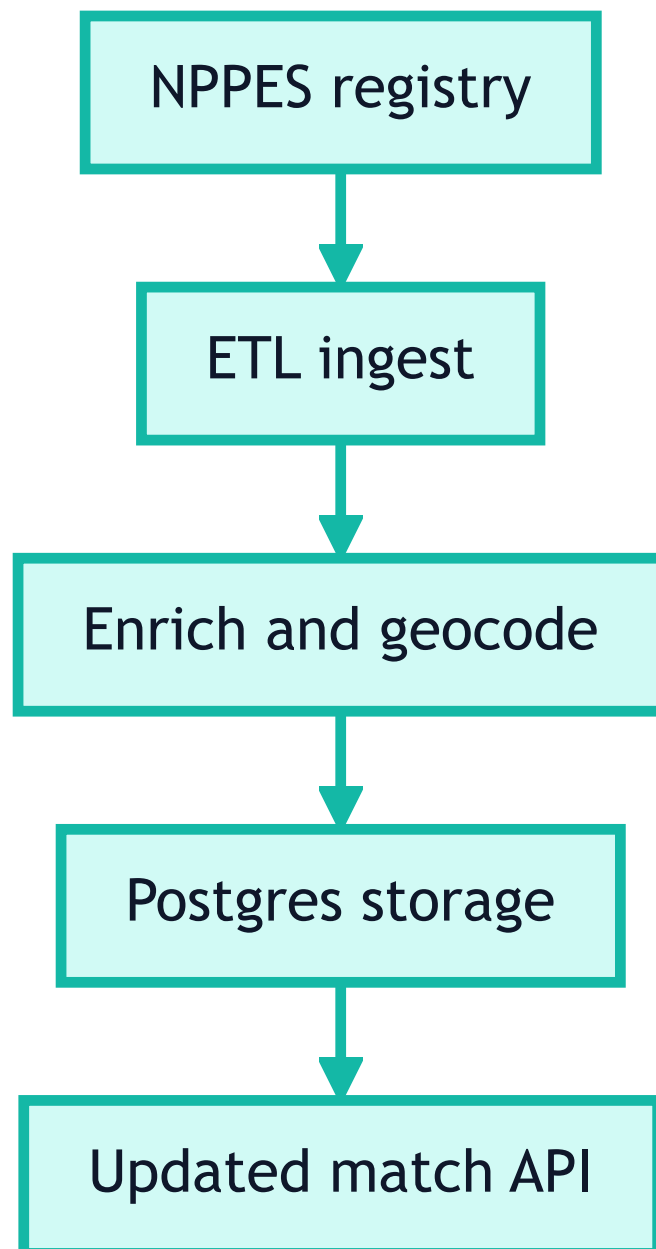
A2. MVP2: Validate and Harden (Days 8–30)

A2.1 Goal

Turn the prototype into a pilot-ready product with real provider data, real users, and the foundation for the first employer contract.

A2.2 Real Provider Data Pipeline

Replace seeded `providers.json` with a dataset of real, contactable providers.



Target: ~500 real providers across 5 metros (Boston, New York, Chicago, Los Angeles, Houston).

Success metric: At least 90% of result cards resolve to a real, contactable provider with a functioning phone number or online booking link.

A2.3 Session Persistence

Deploy managed Postgres with a `sessions` table. Migrate session storage from in-memory map to database-backed adapter; session TTL of 72 hours. Opt-in longitudinal tracking via magic-link authentication with a `completed_intakes` table and a `/history` re-screen view.

A2.4 Clinical Advisory Validation

Engage 2–3 licensed clinicians to review:

- PHQ-9 and GAD-7 item presentation and pacing
- Care-level mapping logic (urgent vs. therapy vs. coaching appropriateness)
- Story-to-draft mapping quality
- Crisis detection sensitivity and false positive rate

Validation results shall be documented in the README as a transparency artifact.

A2.5 Employer Dashboard v0

Aggregate-only employer dashboard supporting the first pilot conversation:

- `employer_accounts` table; domain-based employee access; no individual-level data without consent
- Metrics: completion rate, care-level distribution, time-to-first-match, weekly trends, crisis rate (percentage only)
- Route: `/employer/dashboard`; employer admin auth via magic link or SSO stub

HIPAA note: Aggregate-only data does not constitute PHI. HIPAA-aware architecture review should begin at this phase in anticipation of identified-data pilots.

A2.6 First Employer Pilot Configuration

Target: 1 signed mid-market employer (at least 200 employees).

Pilot setup: Employer admin account; branded entry URL (`/start/<employer-slug>`); ZIP-cluster provider set; dashboard provisioned.

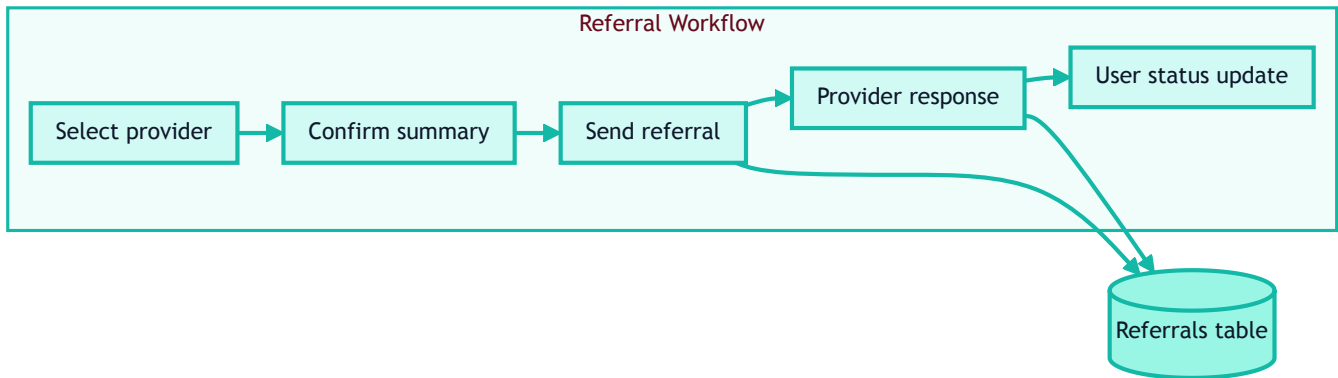
Instrumentation: `intake_events` table (started, profile completed, screening completed, results reached, provider clicked); time-to-first-appointment tracked via follow-up survey.

A2.7 MVP2 Infrastructure Changes

Change	Notes
Add Postgres	Sessions, providers, employers, events
Add provider ETL script	NPPES ingest + normalization; runs on schedule
Add employer auth	NextAuth.js or Auth.js (magic link)
Add aggregate dashboard route	<code>/employer/dashboard</code>
Update <code>/api/match</code>	Read from database instead of JSON file
Retain scoring/safety architecture	No changes to deterministic scoring or crisis logic

A3. MVP3: Close the Loop and Scale (Months 2–4)

A3.1 Provider Notification and Closed-Loop Referral



This closed loop enables Phase 2 monetization (per-completed-referral fee to insurer), time-to-first-appointment measurement against the 48-day baseline, and provider fit quality measurement.

A3.2 Live Insurance Eligibility Verification

Integrate with a clearinghouse (e.g., Availity, Change Healthcare). Inputs: carrier, optional member ID, provider NPI. Outputs: in-network status and estimated out-of-pocket. Display on results with a fallback to verify directly with the provider.

Compliance: Eligibility queries require PHI handling and a Business Associate Agreement with the clearinghouse.

A3.3 Booking Handoff Options

Option	Description
A	Deep-link to provider online scheduling (public URL)
B	Phone scheduling with referral context (care level, non-diagnostic)
C	Calendar integration (e.g., Cal.com) for opted-in providers

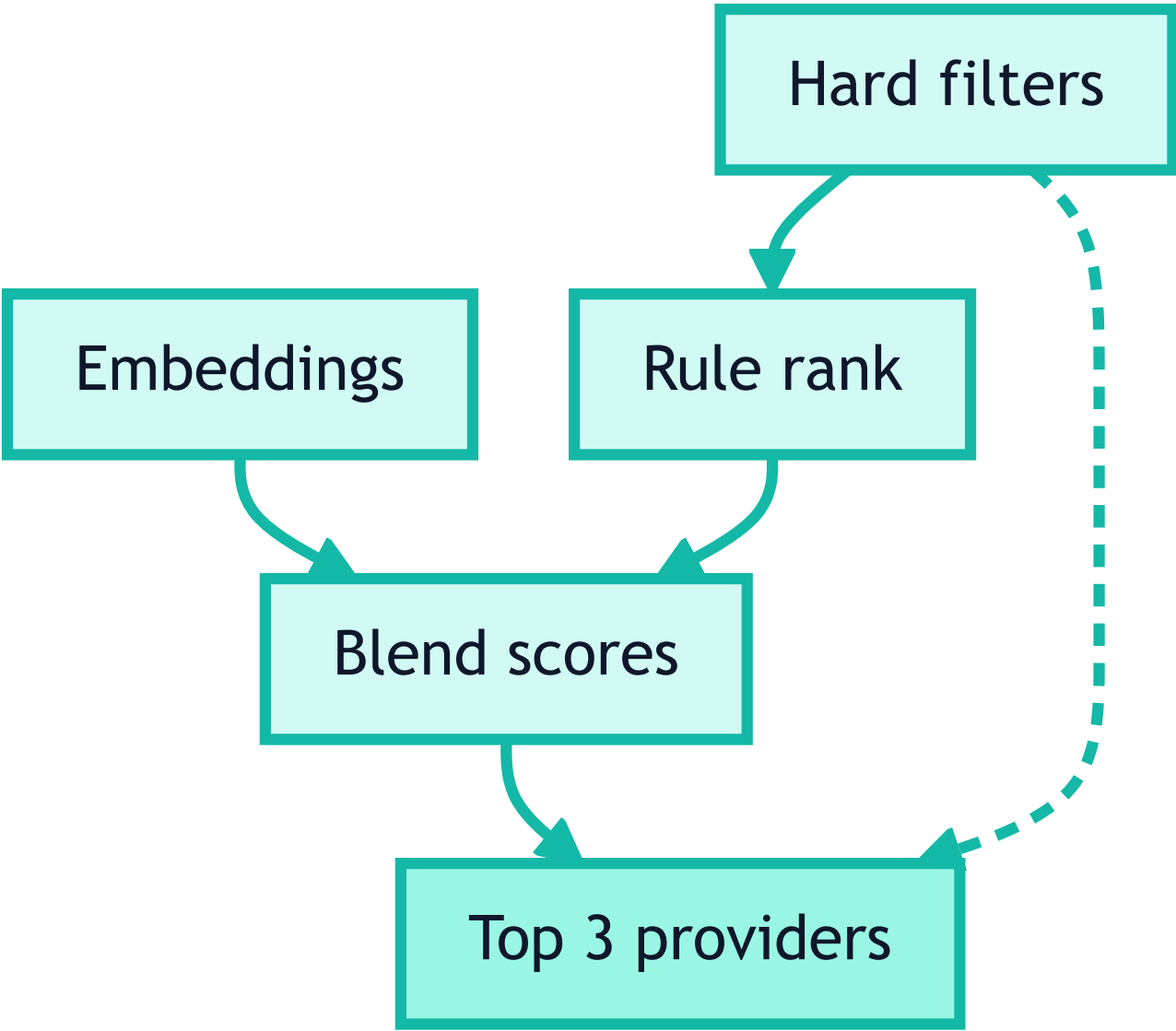
Full EHR integration (FHIR R4 write) is out of scope for MVP3.

A3.4 FastAPI AI Orchestration Service

FastAPI is introduced when AI workloads exceed what Next.js API routes should host: background jobs (ETL, batch re-scoring), embedding computation, model hosting, or automated evaluation on deploy.



A3.5 Embeddings-Based Semantic Matching



A4. Advanced AI Capabilities

A4.1 Adaptive Multi-Instrument Screening

When user concern and PHQ-9/GAD-7 results indicate supplemental assessment would improve routing, additional instruments may be conditionally presented.

Candidate instruments:

Instrument	Items	Scale	Trigger
PCL-5 (PTSD)	20	0–4	Trauma-related concern
AUDIT (alcohol use)	10	Standard	Substance-related concern
MDQ (mood disorder)	13	Standard	Mood concern with mild–moderate PHQ-9

Architecture: Extend the batch intake engine dynamically from ScoreResult and concern; add adaptive instrument logic; extend scoring and matching additively without replacing existing components.

Timeline: MVP3+, subject to clinical advisory board approval.

A4.2 LangGraph Workflow Orchestration

LangGraph shall be evaluated when the intake workflow requires: conditional branching on intermediate scores, tool-calling mid-flow (e.g., eligibility checks), or human-in-the-loop review for crisis cases. The linear batch flow does not require LangGraph in MVP1.

A4.3 Open-Source / Self-Hosted Crisis Classification

Driver	Rationale
Cost	Lower marginal cost at high classification volume
Privacy	Crisis-related text remains on controlled infrastructure
Customization	Model tuned on consented, de-identified examples

Requirements: IRB or equivalent for training data; GPU hosting; precision/recall evaluation with false-negative target under 2%; demographic parity audit.

A5. Enterprise Features

Feature	Description	Phase
SSO (SAML/OIDC)	Enterprise IT requirement for employer access	MVP3+
HRIS integration	Eligibility sync from HR systems (Workday, BambooHR, etc.)	MVP3+
EAP platform integration	Single benefits surface; ClearPath accessible within existing EAP	MVP3+
Role-based access + audit logs	Security and compliance; admin roles, clinician reviewer roles	MVP3+
White-label / co-branding	Custom domain, logo, color scheme for channel partners	Scale
Configurable care pathways	Per-employer routing rules	Scale
BAA + HIPAA-compliant data plane	Required before any employer pilot involving identified employee health data	Before identified-data pilot
Multi-language (Spanish first)	Intake instruments have validated Spanish translations (PHQ-9-ES, GAD-7-ES)	MVP3+

A6. Architecture Evolution

Stage	Architecture	Driver
MVP1 (current)	Single Next.js app; seeded JSON; in-memory sessions; Vercel	Minimal infrastructure; validate core loop
MVP2	Next.js + managed Postgres; provider ETL; opt-in session store; employer auth	Real data; persistence; pilot readiness
MVP3	Add FastAPI AI service; pgvector for embeddings; queue for background jobs	AI orchestration; async work; semantic matching
Scale	Containerized services; autoscaling; observability; CDN; regional data isolation	Multi-employer load; HIPAA compliance; SLA guarantees

Principle carried throughout all phases: Deterministic scoring remains pure code. Safety resources remain prominent. The crisis page remains static and always-available. PHQ-9/GAD-7 totals never come from unconfirmed AI drafts.

A7. Extensibility Seams

The MVP1 architecture was designed with explicit extension points to allow future work without requiring a rewrite.

Seam	MVP1 Implementation	MVP2+ Extension
Scoring engine	Pure TypeScript; no external dependencies	Add new instruments (PCL-5, AUDIT) as additive score components; interface unchanged
Provider matching engine	JSON file input	Swap to Postgres query + pgvector cosine similarity; interface unchanged
Intake flow engine	Linear 7-batch sequence	Add adaptive branching; add persistence; interface extended, not replaced
Crisis detection module	Rules + API LLM	Add audit log, human review queue, self-hosted classifier; same output interface
LLM abstraction layer	Single provider abstraction	Add provider switching; A/B test models; add streaming support
Provider data source	Static seed file	Replace with database-backed provider service; same ProviderCard output shape
Session storage	In-memory map	Drop-in replacement with database-backed adapter; same interface
Crisis UI component	Static card with links	Add dynamic content (local resource lookup by ZIP); same component contract